

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	25 February 2026
Team ID	699d5f82fdf409fdbd1bec66
Project Name	Malaria Detection using Deep Learning
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Dataset Preparation	USN-1	As a developer, I want to download the malaria dataset from Kaggle so that the model can be trained on real data.	3	High	Arham Pagariya
Sprint-1	Dataset Splitting	USN-2	As a developer, I want to split the dataset into training and validation sets so that model performance can be evaluated.	2	High	Arham Pagariya
Sprint-1	Data Preprocessing	USN-3	As a developer, I want to preprocess images and normalize pixel values so that they can be used by the CNN models.	3	High	Arham Pagariya
Sprint-1	Data Augmentation	USN-4	As a developer, I want to apply image augmentation techniques to improve model generalization.	2	Medium	Arham Pagariya
Sprint-2	Custom CNN Model	USN-5	As a developer, I want to build a custom CNN model to classify malaria-infected cells.	4	High	Anushka Nibandhe

Sprint-2	Transfer Learning	USN-6	As a developer, I want to implement MobileNetV2 and EfficientNetB0 models using transfer learning.	4	High	Anushka Nibandhe
Sprint-2	Model Training	USN-7	As a developer, I want to train all models using the dataset so that predictions can be learned.	3	High	Anushka Nibandhe
Sprint-2	Model Evaluation	USN-8	As a developer, I want to evaluate models using metrics like Accuracy, Precision, Recall, F1-Score and ROC-AUC.	2	Medium	Anushka Nibandhe
Sprint-3	Model Selection	USN-9	As a developer, I want to select the best performing model based on ROC-AUC and accuracy.	2	High	Aricia Dubey
Sprint-3	Prediction Utility	USN-10	As a developer, I want to create functions to preprocess images and predict malaria infection.	2	High	Aricia Dubey
Sprint-3	Flask Backend	USN-11	As a developer, I want to build a Flask backend to serve the trained model for predictions.	3	High	Aricia Dubey
Sprint-3	Image Upload Feature	USN-12	As a user, I want to upload blood smear images to the system to check infection.	2	Medium	Aricia Dubey
Sprint-4	Result Display	USN-13	As a user, I want to see prediction results with probability and class label.	2	Medium	Anvaya Kulkarni
Sprint-4	Testing & Debugging	USN-14	As a developer, I want to test the system to ensure correct predictions.	2	High	Anvaya Kulkarni
Sprint-4	Documentation	USN-15	As a developer, I want to prepare project documentation and reports.	2	Medium	Anvaya Kulkarni

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	10	3 Days	25 Feb 2026	27 Feb 2026	10	27 Feb 2026
Sprint-2	13	4 Days	28 Feb 2026	03 Mar 2026	12	03 Mar 2026
Sprint-3	9	3 Days	04 Mar 2026	06 Mar 2026	8	06 Mar 2026
Sprint-4	8	3 Days	07 Mar 2026	09 Mar 2026	-	09 Mar 2026

Velocity:

Velocity = Story points completed per sprint.

Example:

Sprint-1 completed = 10 points

Sprint duration = 3 days

$$Velocity = \frac{10}{3}$$

$$Velocity \approx 3.3 \text{ story points/day}$$

So, the team velocity $\approx$ 3 story points per day.

Burndown Chart:

A Burndown Chart is a graphical representation of the amount of work remaining in a project over time. It helps the development team track progress and determine whether the project is on schedule. In Agile methodology, it shows how quickly the team is completing tasks during each sprint.

In the Malaria Detection System project, the burndown chart illustrates the reduction of story points as tasks such as dataset preparation, model training, system integration, and testing are completed.

